

Agentic AI Voice Bot

Built on the LiveKit Framework

Overview

This document describes the design and feature set of an Agentic AI Voice Bot built with the LiveKit framework (github.com/livekit). The bot lets users talk naturally to manage Google Notes, Google Calendar meetings, and Gmail. It also handles voice interruptions in real time so conversations feel smooth and natural.

1. Google Notes

Create a Note

The user speaks the note content aloud. The bot calls the Google Keep API and saves it immediately.

Edit a Note

The user names the note and speaks the new content. The bot finds the note and updates it.

Read All Notes

The bot fetches the full list and reads back each note title with a short preview of its content.

Delete a Note

The user names the note to remove. The bot confirms the action and then deletes it.

2. Google Calendar

Create a Meeting

The user gives the title, date, time, and guests by voice. The bot creates the event and sends invites.

Edit a Meeting

The user names the meeting and states what to change. The bot updates the calendar event accordingly.

Fetch All Meetings

The bot reads back upcoming events including date, time, and guest list for each one.

Delete a Meeting

The user names the meeting to remove. The bot confirms and deletes it from the calendar.

3. Gmail

Read All Emails

The bot fetches recent emails and reads the sender, subject, and a short preview of each message.

Create a Draft

The user dictates the recipient, subject, and body. The bot saves it as a Gmail draft.

4. Voice Interruption

Real-Time Stop

If the user speaks while the bot is talking, the bot stops its audio output right away.

Active Listening

The bot switches to listening mode and fully processes the new voice input.

Fresh Response

The bot replies to the new query only, without mixing it with the previous response.

Technology Stack

Voice Transport

LiveKit Agents SDK using WebRTC for real-time audio streaming

Speech to Text

Deepgram or OpenAI Whisper via LiveKit STT plugin

Language Model

OpenAI GPT-4o or Anthropic Claude via LiveKit LLM plugin

Text to Speech

ElevenLabs or OpenAI TTS via LiveKit TTS plugin

Google Notes

Google Keep API

Google Calendar

Google Calendar API v3

Gmail

Gmail API v1

Interruption

LiveKit Voice Activity Detection with real-time turn detection

Backend

Python with the LiveKit Agents framework

How It Works

Step 1 User speaks

Microphone audio is streamed to LiveKit in real time.

Step 2 Speech to text

LiveKit transcribes the audio using Deepgram or Whisper.

Step 3 Intent detection

The LLM reads the transcript and decides which tool or action to call.

Step 4 Tool execution

The agent calls the correct Google API and retrieves the result.

Step 5 Voice reply

The result is converted to speech and played back to the user.

Step 6 Interruption check

If the user speaks during playback, the bot stops and returns to Step 1.

Key Design Decisions

- LiveKit handles all audio transport so the bot works over any network with low delay.
- Tool calls are defined as functions so the LLM can pick and run them without extra logic.
- Voice Activity Detection runs alongside playback so interruptions are caught within 200ms.
- Google OAuth tokens are stored server-side and refreshed automatically to keep sessions active.
- All Google API calls include error handling so failures are reported to the user by voice.
- The agent keeps a short session memory so follow-up questions work without repeating context.

References

- LiveKit Agents SDK: github.com/livekit/agents
- LiveKit Documentation: docs.livekit.io
- Google Keep API: developers.google.com/keep
- Google Calendar API: developers.google.com/calendar
- Gmail API: developers.google.com/gmail/api